Society of Petroleum Engineers

SPE-226627-MS

UAEs Unconventional Shale Gas Play Appraisal Technical and Operational Learnings

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This paper was prepared for presentation at the SPE International Hydraulic Fracturing Technology Conference and Exhibition held in Muscat, Sultanate of Oman, 23 - 25 September 2025.

Abstract

The paper presents the technical and operational insights gained throughout the appraisal program, detailing the process from wellbore preparation to production. It explains how intervention tools and practices differed between two wells drilled within the same target, just 500 feet apart. The discussion also covers strategies to maximize operational efficiency, reduce emissions, and leverage local resources, as well as the adaptation and customization of technologies and tools tailored to the UAE's unconventional formations.

Multistage well completions have evolved significantly over the last two decades, driven largely by the technological revolution in unconventional shale oil and gas development. Innovations in tools, techniques, and concepts have enabled the efficient exploitation of ultralow-permeability resources on a large scale. However, these methods must be carefully evaluated for early-life fields, where reservoir properties and stress regimes can differ substantially. If not properly designed, hydraulic fracture treatments may be ineffective and jeopardize project outcomes, with implications for national energy self-sufficiency and strategy.

The Diyab Reservoir, an organic-rich limestone unconventional play, is located in the Al Dhafra region, about 250 km west of Abu Dhabi. The target formations are characterized by extremely high-stress rock, as well as high-pressure and high-temperature (HPHT) conditions.

The appraisal strategy involved completing two horizontal wells on a pad simultaneously (zipper fracturing) with extensive multistage hydraulic fracture treatments using the perf-and-plug method. Both wells underwent long-term flow testing through an early production facility to minimize emissions while gathering critical data to address uncertainties and design a robust field development plan.

Several challenges have underscored the importance of multidisciplinary integration and planning, wellbore construction impacts, contractor performance, and tool reliability. This paper summarizes case studies and operational results from the first horizontal well interventions in UAE's HPHT reservoirs, providing important contributions and key insights for achieving UAE's gas self-sufficiency and supporting its energy transition strategy.

Introduction

DIYAB Regional Context

The Upper Jurassic Diyab Formation is the source rock for several world-class oil and gas fields in the Middle East and in recent years its potential as an unconventional play was evaluated in the region (United Arab Emirates (UAE), Saudi Arabia, Kuwait, and Bahrain) **(Fig.1)**.

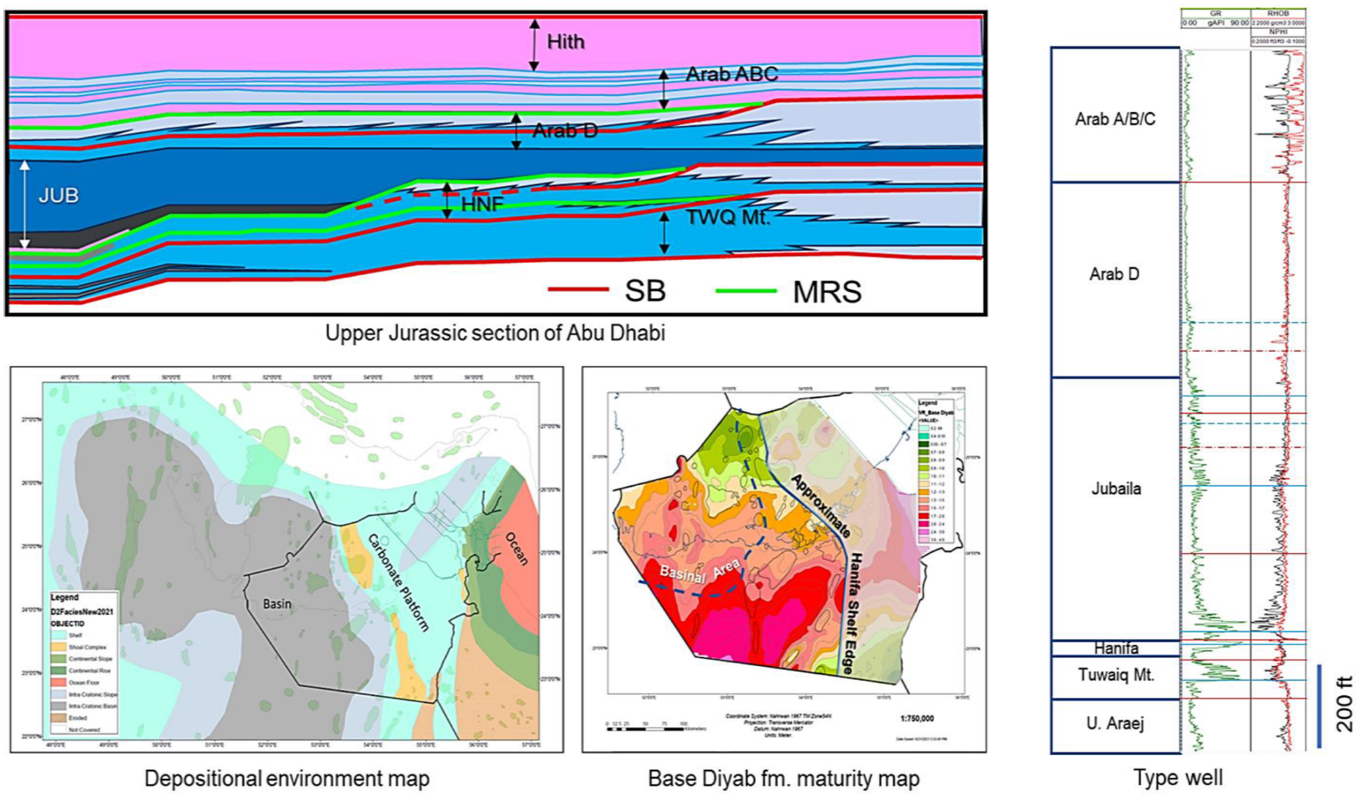

Figure 1—Abu Dhabi Diyab fm. geological overview; type well, Upper Jurassic section of Abu Dhabi, depositional environment map and maturity map from Van Laer, et al (2022).

The extent of the onshore prospective Diyab interest zone (composed of Jubaila, Hanifa and Tuwaiq Mountain formations) within the UAE covers 25,000 km^2.

Diyab formation in the UAE lies approximately 2,000 feet deeper than comparable formations in Saudi Arabia, resulting in heightened reservoir temperatures and greater thermal maturity. This geological uniqueness is further accentuated by the formation's pronounced carbonate mineralogy and elevated pressure gradient, distinguishing it from conventional "shale" plays. The predominance of carbonate content contributes to the brittleness of the reservoir, making it especially receptive to hydraulic fracture stimulation. When coupled with the high-pressure gradient, Diyab emerges as a compelling new exploration frontier, supporting the development of innovative extraction techniques and strategies.

Within the Diyab, various organic-rich intervals are interspersed with less organic, tight limestone layers, effectively creating distinct flow units and offering multiple zones for potential hydrocarbon recovery. This stratified nature enables the possibility of stacked play development, allowing for simultaneous exploitation of several resource horizons and maximizing field productivity.

From a project planning perspective, the region presents opportunities for an expansive drilling campaign, with projections exceeding 2,000 wells. The scale and complexity of operations necessitate robust logistical coordination, advanced reservoir modeling, and adaptive completion practices tailored to the demanding high-pressure, high-temperature (HPHT) conditions. Technical teams have prioritized the integration of real-time monitoring systems and collaborative evaluation approaches to address geological uncertainties, optimize fracture placement, and ensure sustainable production.

These ongoing initiatives not only contribute to the successful appraisal and development of the Diyab Formation but also reinforce the UAE's commitment to energy self-sufficiency, technological innovation, and strategic resource stewardship in the context of its broader energy transition objectives.

Appraisal Program

The Initial Exploration and appraisal program focused on Block 1 in partnership with Total energies after an agreement signed end of 2018 between the two companies.

The appraisal program focused on the Hanifa formation, where two appraisal wells were drilled and tested. An additional two step-out exploration wells were drilled and tested in the same bench, to assess the paly extents.

The program objective was to evaluate the repeatability and impact of longer lateral length and higher proppant loading on well productivity. A six-month long-term test was performed to validate type curve parameters through an early production facility which was connected the produced gas pipeline to sales station.

Operations Learnings

Over the past years, the UAE's unconventional operations have undergone a significant learning curve, drawing valuable lessons from both rig and rigless activities. This process of continual refinement has fostered a strong culture of innovation and adaptability within project teams, who have worked collaboratively to overcome geological and operational challenges unique to the Diyab Formation.

A primary area of focus has been the optimization of drilling and completion strategies under high-pressure, high-temperature (HPHT) conditions. Teams have leveraged real-time data acquisition and advanced geosteering technologies to precisely target organic-rich intervals and minimize non-productive time. Additionally, experience with multi-stage hydraulic fracturing in the brittle carbonate-rich rock has led to the adoption of tailored proppant selection, fluid systems, and stage spacing, resulting in improved fracture conductivity and well productivity.

The operational learning extends to logistics and supply chain management as well, ensuring that critical materials and equipment are delivered efficiently despite the remote and challenging desert environment. Health, safety, and environmental (HSE) protocols have been strengthened through continuous risk assessments, preventative maintenance schedules, and rigorous emergency response planning.

Moreover, collaboration with technology partners and service providers has accelerated the adoption of fit-for-purpose tools—such as high-strength casing designs, advanced packers, and zonal isolation technologies—further enhancing well integrity and longevity. Knowledge gained from early wells has been systematically documented and integrated into subsequent project phases, fostering a data-driven approach to decision-making and continuous improvement.

By adapting strategies and pioneering tailored solutions, the team has been able to optimize performance, streamline processes, and lay a solid foundation for future development in the challenging context of the Diyab Formation. These cumulative experiences and innovations not only drive technical success but also support cost efficiency and sustainable resource development for the UAE's energy sector.

Long Lateral Well Architecture

Well Architecture Evolution. The operation efficiency is maximized through the well architecture design. **Fig.2** shows well architecture evolution along the exploration and appraisal phases which supports the well integrity objectives and reduces the operation duration in the long lateral wells.

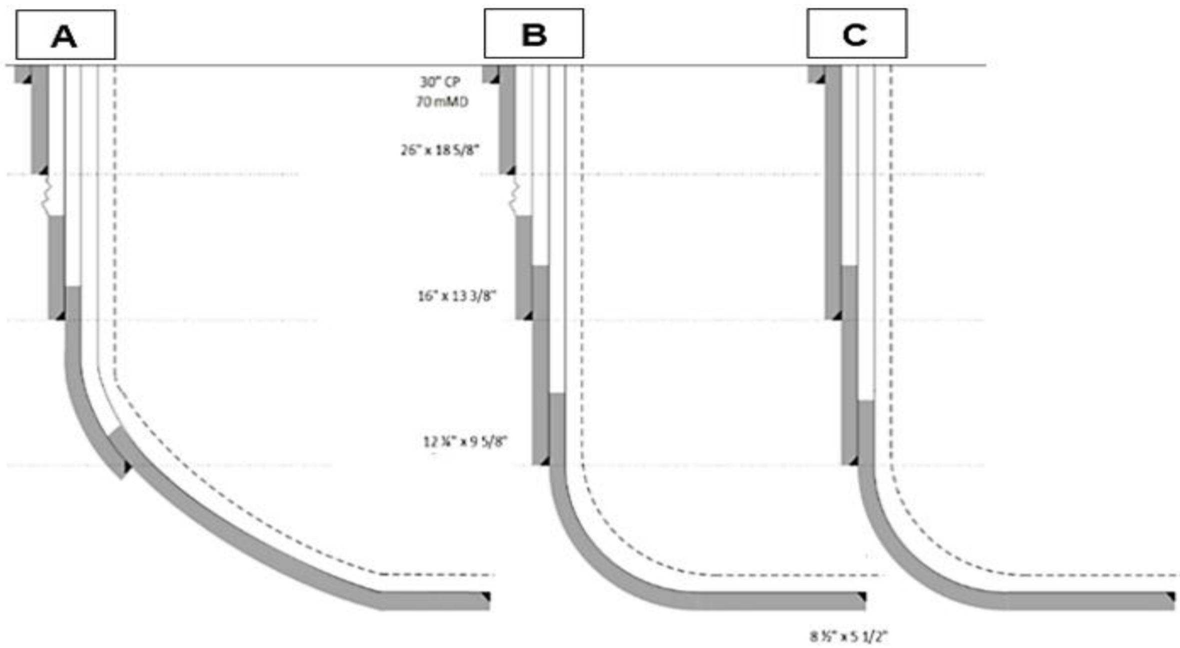

Figure 2—Well Architecture Evolution

Fig.2 shows:

- - **(A)** Well design shows an old well architecture with shallow kick off point (KOP) and shorter lateral.
- - **(B)** Well design:
  - 12 1/4" section drilled vertically, better performance + shorter well.
  - 8 ½" section curve with more aggressive build up making the wells shorter for same horizontal section.
- - **(C)**well design changes from design **(B):**
  - Dropping of 18-5/8" section which involve no more 26" drilling and 18 5/8" casing operation and saved 400 m of casing cost per well.

Fracturing Fluids

Treatment fluids were tested in Diyab formation and optimized to carry larger proppant concentrations without causing any damages to Diyab reservoir.

HVFR used in UAE unconventional formation to reduce the logistics cost **(Fig. 3),** improve environmental spills by eliminating petroleum distillates and flexibility in real time adjustment based on the formation pressure response.

HVFR is a copolymer powder that hydrates rapidly into solution on contact with water at 40°F or above to enable hydraulic fracturing.

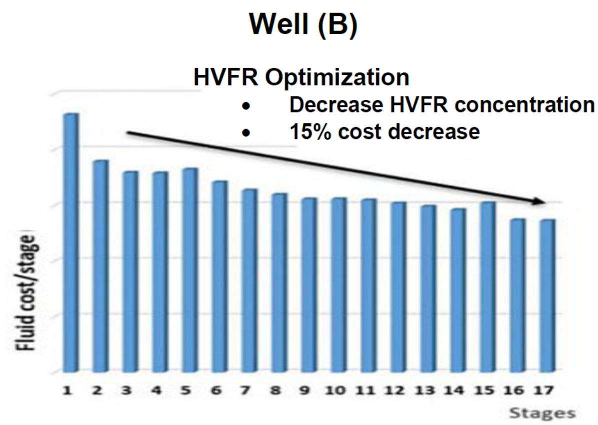

Figure 3—HVFR Optimization in the first 17 stages

HCl is a Hydrochloric acid used in a fracturing fluid, to enhance the frac propagation.

Well **(B)** and **(C)** are targeting the same formation (Hanifa): **Fig.3** & **Fig.4** are showing HVFR concentration and HCL acid cost reduction per stage.

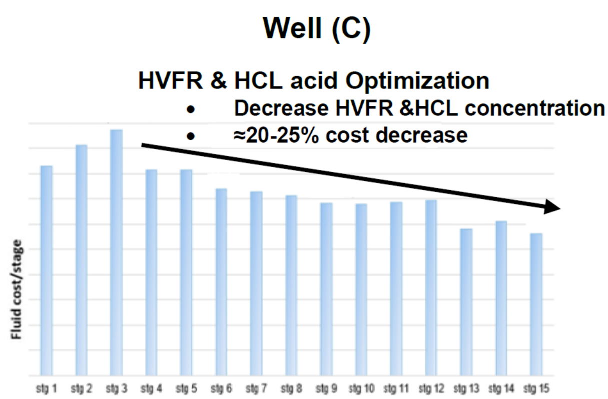

Figure 4—HVFR & HCL acid Optimization in the first 15 stages

At low concentrations, HVFR can be used as a friction reducer for high-rate slick water treatments. At high loadings, HVFR generates higher viscosities than conventional friction reducers, improving proppant transport. It also generates higher viscosity compared to same polymer loading of guar gel **(Fig.).**

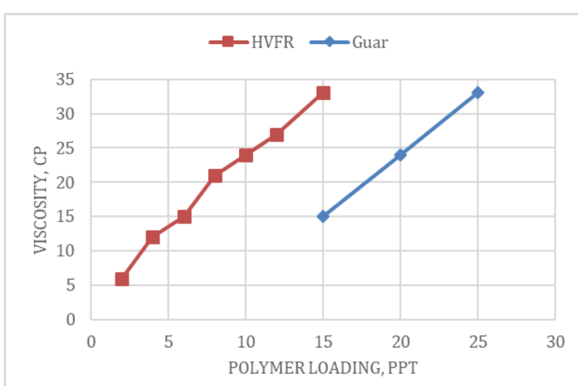

Figure 5—Viscosity comparison between HVFR and guar gel at 170 S^{-1} and 70°F with treatment water.

Proppant Selection and Fracture Geometry

Low-density high-strength proppant (HSP-LD) was selected as the main proppant for its cost, and low concern over crushing due to the managed choke flow back approach (**Table 1** & **Fig. 4**).

Table 1—Comparison between HSP-LD and HSPs

Proppant type	Crust at 12 500 psi (%)	Conductivity at 10,000 psi(md-ft)	Permeability at 10,000 psi (D)
SLB 40/70 HSP LOW DENSITY	3.76	780	43
SLB 40/70 HSP NORMAL	4.46 (crush at 15,000 psi)	733	56
Carbo 40/70 HSP NORMAL	1.4	1000	75

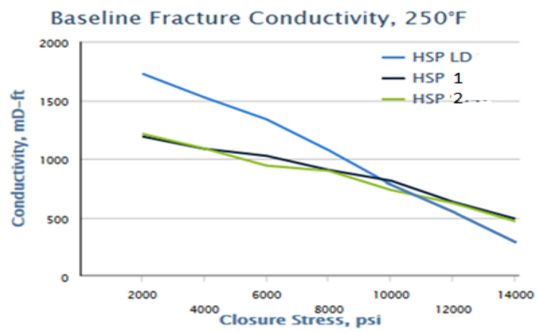

Figure 6—Fracture conductivity comparison between HSP-LD and HSPs

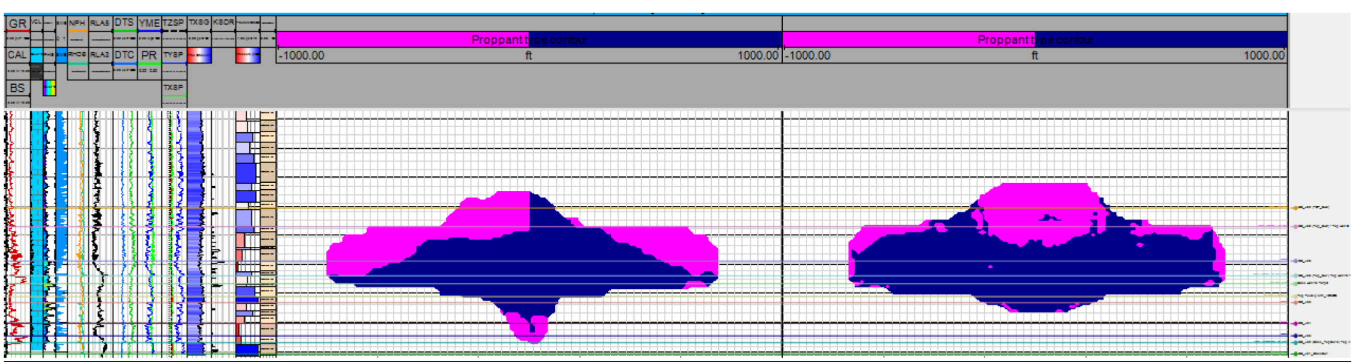

Figure 7—Fracture geometry with proppant type contour (purple, 100 mesh; dark blue, 40/70 mesh) comparison for standard HSP (S.G 3.7) and HSP-LD (S.G. 2.7), 40/70 mesh. HSP-LD gives better longer propped area. From Pourpak, H., Newby, W., Taubert, S., et al. 2021.

Plug and Perf Completion

Dissolvable plugs (**Fig.8**) were mainly selected to eliminate the need for milling. The use of dissolvable plugs enables restoring the treatment path if the plug is stuck at any restriction point caused by casing deformation and/or proppant accumulation. Furthermore, slim dissolvable plugs were prepared in case severe casing deformation occurred that would stop the standard version of dissolvable plugs passing through deformed section.

Figure 8—Dissolvable Plug

The Use of dissolvable plugs requires a more intricate operation sequence arrangement because the working time of this dissolvable plug is expected to be approximately 5 to 6 hours at bottomhole conditions (330°F and low salinity fracturing fluids).

Pumping the cooling fluid into the well bore increased the lifetime of the dissolvable plug and it is perfectly worked in the zipper frac operations.

Well Completion Technologies

A selection of tools and BHA were available with all possible contingencies, with enough backup to address any intervention complication, lesson learned from other ADNOC wells completed in areas, wireline BHA and equipment were modified to address the high shut-in pressure post frac, pump down procedures were revised based on the learning from the initial phase and high-density brine was made available to lower pressure in case of difficult well interventions.

Green completion

H2S Sweeting closed loop

H2Ss presence during early flow back and closed loop operation was addressed by the implementation of the H2S scavenging scrubber system which eliminated the use of harsh scavenger, reduced footprint (less tanks required) and cleaner treated waster with no chemical that can be disposed off without expensive filtration or treatment.

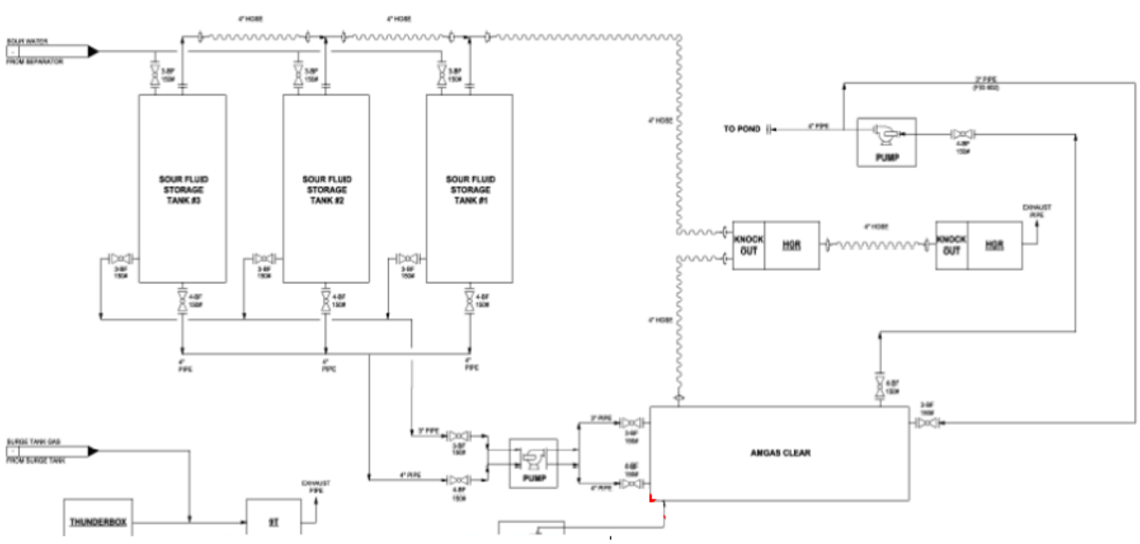

Figure 9—H2S scavenging scrubber system

H2S scavenging scrubber system design expanded to treat a huge volume of flow back water in the zipper frac operation by increasing the number of storage tanks and adding.

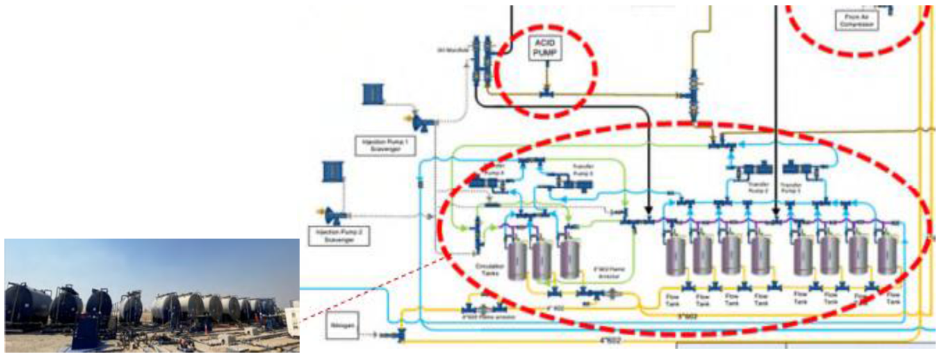

Figure 10—H2S scavenging scrubber system in the zipper frac Operation

Green House Gas (GHG) Emissions Reduction

Gas flaring was reduced by connecting the wells to an early production facility, wells were long term tested and gas was exported, a low emission unconventional gas was sold in less than 2 years from discovery a record in the region (**Fig.11**).

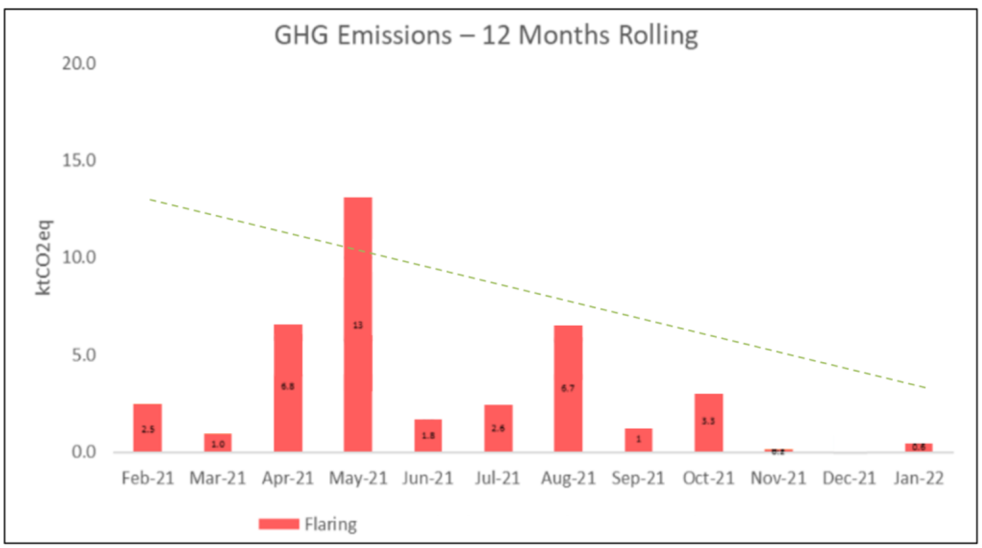

Figure 11—GHG Emissions Reduction

Zipper Frac

Frac Surface Setup

The location setup focused on enabling fracturing treatment for two wells with 100 m of surface spacing that could be pumped alternatively and continuously. Besides subsurface factors affecting operation efficiency, the zipper fracturing manifold and water management are the key points to address. Split-stream operation was not planned to reduce the complexity for this first-time zipper fracturing operation in the country.

Fig.12 shows the overall well-site layout, with details explained in the subsections below.

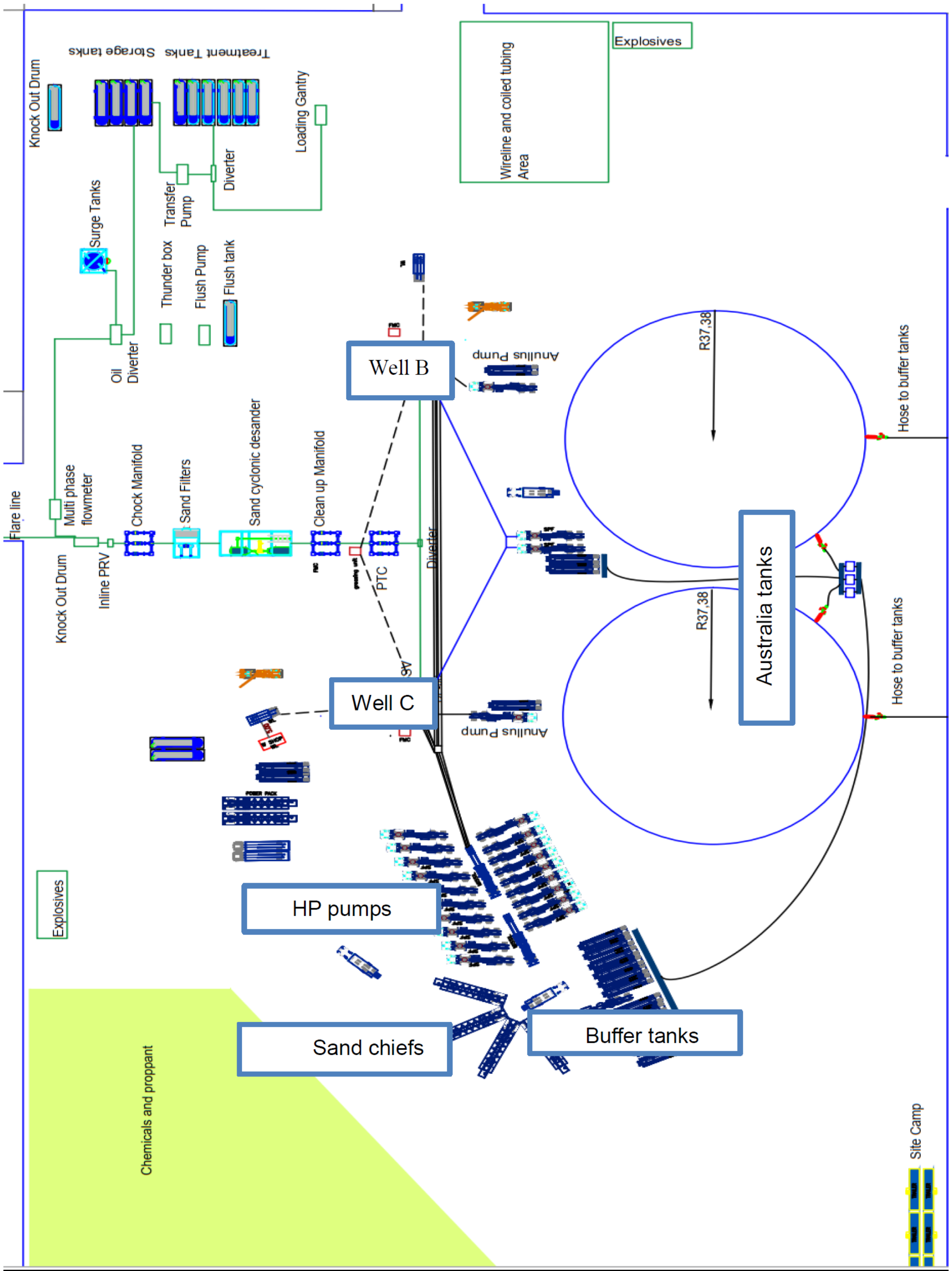

Figure 12—Wellsite layout

Water Management

Two Australian tanks enabled a continuous fracturing operation, with 80,000 bbl of volume each and 12 buffer tanks with 500 bbl of volume each were equipped onsite (Fig. 12). Furthermore, there were 15 water tanks installed to buffer the water loaded from water haulers **(Fig. 13).** Considering roughly 13,000 bbl of fluid consumption per stage, the water onsite can support a maximum 12 stages. The daily water supply could reach maximum 30,000 bbl per day if required.

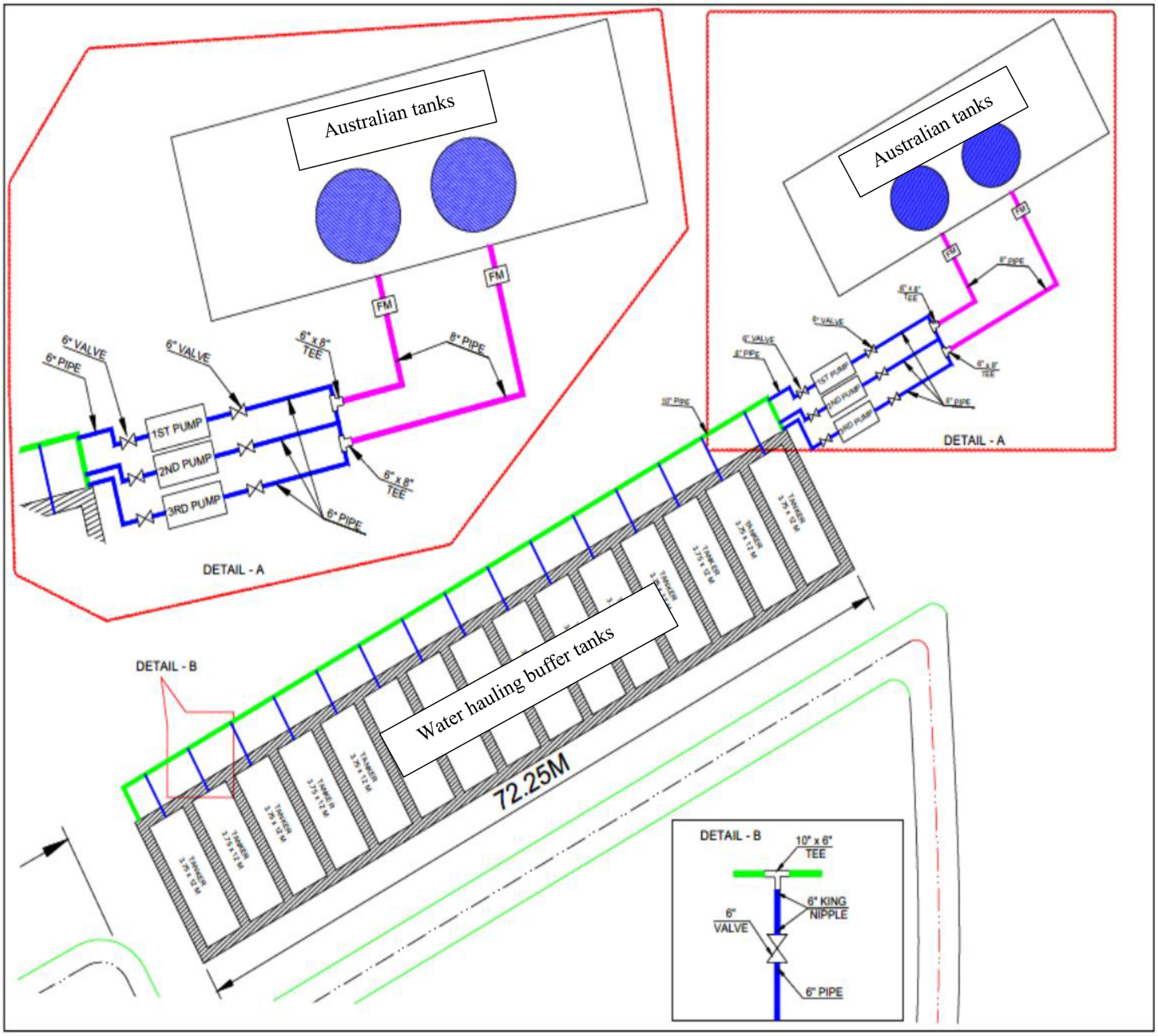

Figure 13—Water supply to Australian tanks

Zipper frac Operation Efficiency

Operational efficiency was affected by unexpected events, but the set up allowed to continue operation on one well while the second well was being addressed.

Fig. 14 summarizes the time in hours for each stage of fracturing treatment and wireline pump down operations.

Figure 14—Summary on fracturing and wireline operation hours for all stages for Well B (top) and Well C (bottom)

Execution Results

Despite the challenges from complex geology, high tectonic stress, casing deformation, remote location and pandemic lockdown treatment were successfully completed, and contingencies planning helped in reducing NPT due to failure and well intervention complexity.

Once well clean up early frac load and to avoid flaring gas the pad was immediately connected to the early production facility where the well was tested for a period exceeding six months, without excessive flaring making it a green completion and low emission gas production. Learning was captured during this phase and will be implemented during the upcoming development phase.

Conclusion

This paper has detailed the technical and operational journey of completing the appraisal program for two wells within the same target formation, from wellbore preparation to the production phase. The focus was on maximizing operational efficiency, minimizing emissions, and leveraging local resources to enhance project outcomes.

Key Learnings and Achievements

- Well Architecture Optimization: Eliminating the 18-5/8" casing section and associated 26" drilling reduced casing and drilling costs.

- Frac Fluid Design: Optimized fluid design lowered logistics costs, reduced environmental risks by eliminating petroleum distillates, and allowed real-time adjustments based on reservoir response.
- Proppant Selection: Adoption of low-density, high-strength proppant (HSP-LD) maximized proppant loading and improved cost efficiency within the fracture geometry.
- Completion Tools: Implementation of dissolvable plugs minimized operational time and increased fracturing efficiency.
- H2S Sweeting Closed Loop System: This system eliminated the need for harsh scavengers, reduced surface footprint, and generated cleaner produced water suitable for disposal without costly treatment.
- Zipper Frac Surface Setup and Water Management: The use of large Australian tanks enabled seamless, continuous fracturing operations and efficient water supply management.

Project Impact

- Successful execution of a multi-stage zipper fracturing operation established a new benchmark for hydraulic fracturing in the region, demonstrating the feasibility of high-stage completions with maximum proppant loading.
- Operational and environmental efficiencies achieved through robust planning and adaptive strategies.

These learnings and advancements not only contributed to the immediate success of the appraisal program but also provided valuable guidance for future unconventional resource developments.

Way Forward for Unconventional Resource Operations

Continuous Improvement and Innovation

Looking ahead, there is significant potential for further advancements in unconventional resource operations. Several key considerations can serve as valuable guidance for future well programs:

Best Practices and Recommendations

1. Continuous Monitoring and Real-Time Adjustments:

   Ongoing monitoring, calibration, and real-time design adjustments are essential, especially early in play development. Integrating subsurface data with seismic information is crucial for successful shale gas operations, allowing teams to respond proactively to dynamic reservoir conditions and optimize field performance.
2. Mitigation Planning for Casing Deformation:

   In high tectonic environments, preparing a robust mitigation plan for casing deformation is critical. The use of slim dissolvable plugs has proven to be a practical and efficient solution, minimizing operational delays and ensuring the continuity of fracture treatments.
3. Evaluating Water and Proppant Loading:

   While higher water and proppant loading strategies were implemented, their full impact could not be conclusively validated with these wells. Further studies and data collection are recommended to better understand the relationships between fluid and proppant volumes and ultimate well productivity.
4. Advanced Diagnostic Techniques:

   The application of microseismic monitoring and other diagnostic methods can deliver significant added value. These advanced techniques can help further refine completion designs, inform field development planning, and contribute to maximizing resource recovery while maintaining operational safety and efficiency.
5. Contingency Planning and Communication:

Proper contingency planning and the establishment of clear, consistent communication protocols among all stakeholders are fundamental for efficient and successful execution. Close collaboration and timely sharing of information reduce non-productive time, enhance decision-making, and foster a culture of continuous improvement throughout the project lifecycle.

Acknowledgements

The authors of this paper would like to thank ADNOC and Total Energies for the permission to publish this paper.

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